INCH-POUND

MS9954A
w/AMENDMENT 3
24 April 2024
SUPERSEDING
MS9954A
w/AMENDMENT 2
19 April 2022

DETAIL SPECIFICATION SHEET

PLUG, ALUMINUM, MACHINE THREAD-DRILLED, PREFORMED PACKING

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.

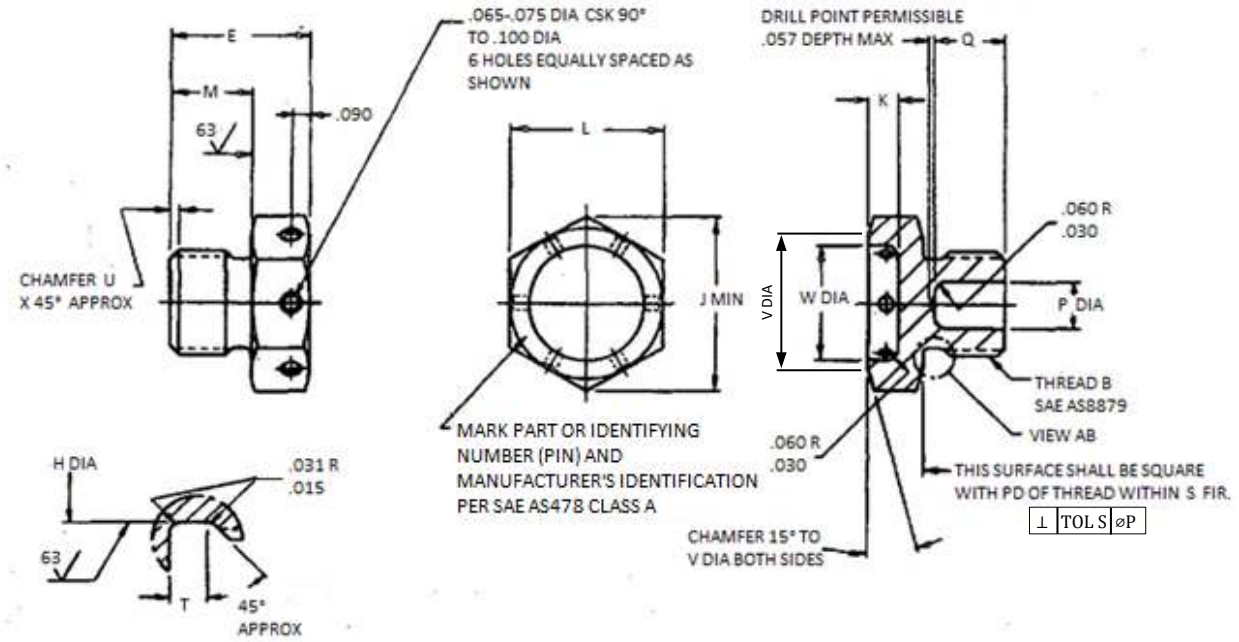


FIGURE 1. Plug (see notes on page 3).

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TABLE I. Dimensions (see notes on page 3).

Part or Identifying Number (PIN).	Thread B (Per ASME B1.1)	E	H +.002 -.003	J min	K	L	M	P	Q	S	T +.015 -.000	U	V	W	Approx. weight lb/100
MS9954-01	.250 -28 UNJF-3A	.560	.196	.482	.150	.430-.439	.320	-	-	.003	.063	.020-.050	.438	.250	.421
MS9954-02	.3125 -24 UNJF-3A	.600	.250	.622	.150	.553-.564	.360	-	-	.003	.063	.020-.050	.562	.312	.731
MS9954-03	.375 -24 UNJF-3A	.600	.312	.694	.150	.616-.627	.360	-	-	.003	.063	.020-.050	.625	.375	.942
MS9954-04	.4375 -20 UNJF-3A	.660	.364	.766	.150	.679-.690	.400	-	-	.003	.075	.030-.060	.688	.438	1.301
MS9954-05	.500 -20 UNJF-3A	.670	.426	.837	.150	.741-.752	.400	-	-	.003	.075	.030-.060	.750	.500	1.654
MS9954-06	.5625 -18 UNJF-3A	.680	.481	.907	.150	.803-.814	.400	-	-	.003	.083	.040-.070	.812	.562	2.021
MS9954-07	.625 -18 UNJF-3A	.730	.544	.978	.160	.865-.877	.440	-	-	.003	.083	.040-.070	.875	.625	2.554
MS9954-08	.750 -16 UNJF-3A	.730	.660	1.121	.160	.990-1.002	.440	.250	.400	.003	.094	.040-.070	1.000	.750	3.265
MS9954-09	.8125 -16 UNJ-3A	.740	.722	1.191	.160	1.052-1.064	.440	.312	.400	.003	.094	.040-.070	1.062	.812	3.769
MS9954-10	.875 -14 UNJF-3A	.740	.773	1.263	.160	1.115-1.127	.440	.375	.400	.003	.107	.050-.080	1.125	.875	4.122
MS9954-11	1.000 -12 UNJF-3A	.780	.882	1.475	.190	1.301-1.314	.440	.500	.400	.005	.125	.050-.080	1.312	1.000	5.792
MS9954-12	1.0625 -12 UNJ-3A	.830	.945	1.475	.200	1.301-1.314	.480	.562	.400	.005	.125	.050-.080	1.312	1.000	6.307
MS9954-14	1.1875 -12 UNJ-3A	.830	1.070	1.619	.200	1.427-1.440	.480	.688	.400	.005	.125	.050-.080	1.438	1.125	7.388
MS9954-16	1.3125 -12 UNJ-3A	.840	1.195	1.833	.200	1.614-1.627	.480	.812	.400	.005	.125	.050-.080	1.625	1.312	9.243
MS9954-18	1.500 -12 UNJF-3A	.840	1.382	1.974	.200	1.738-1.752	.480	1.000	.400	.005	.125	.050-.080	1.750	1.438	10.841
MS9954-20	1.625 -12 UNJ-3A	.850	1.507	2.117	.200	1.863-1.877	.480	1.125	.400	.005	.125	.050-.080	1.875	1.562	12.556
MS9954-24	1.875 -12 UNJ-3A	.860	1.756	2.402	.200	2.113-2.127	.480	1.375	.400	.005	.125	.050-.080	2.125	1.812	16.059
MS9954-28	2.250 -12 UNJ-3A	.870	2.131	2.830	.200	2.488-2.502	.480	1.750	.400	.005	.125	.050-.080	2.500	2.188	21.965
MS9954-32	2.500 -12 UNJ-3A	.870	2.381	3.115	.200	2.738-2.752	.480	2.000	.400	.005	.125	.050-.080	2.750	2.438	26.026

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NOTES:

1. Hexagon and diameter V shall be concentric with thread PD within .010 FIR.
2. Diameter H shall be concentric with thread PD within .005 FIR, and free from thread tool marks.
3. Finish: Anodic treatment per MIL-PRF-8625 Type IC, II or IIB, Class 1. Hexavalent chromium is not permitted for anodizing or sealing. Do not dye.
4. Surface texture: ASME B46.1 Unless otherwise specified, surfaces shall be 125 micro inches.
5. Fluorescent penetrant inspection conforming to ASTM E1417/E1417M, Type 1, Sensitivity Level 2. Acceptance in accordance with SAE AS3071.
6. Break sharp edges .003-.015 unless otherwise specified.
7. Dimension in inches. Unless otherwise specified: tolerances; linear dimension ±0.010, angular dimensions ±0.5º.
8. Do not use unassigned part numbers.

Requirements:

Material and material designators:

Blank - Aluminum alloys 2014-T6 or 2014-T651 conforming to SAE AMS4121 or ASTM B211/B211M.
W - Aluminum alloys 7075-T73 or 7075-T7351 conforming to SAE AMS4124 or ASTM B211/B211M.

PIN examples:

MS9954-08 - Specifies a plug aluminum alloys 2014-T6 or 2014-T651 with .750 – 16UNJF-3A threads.
MS9954-08W - Specifies a plug aluminum alloys 7075-T73 or 7075-T7351 with .750 – 16UNJF-3A threads.

Due to stress corrosion cracking, aluminum alloy 2014 “Blank” designator is inactive for new design. For new design use, aluminum alloy 7075 “W” designator.

NOTE: The “W” designator will replace “Blank” designators, however the “Blank” designators will not replace the “W” designators.

NOTE: Parts made between 12 February 2019 and 2 June 2020 shall be accepted as aluminum alloy 7075 “W” designator.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

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Referenced documents. This document references the following:

ASME B46.1	MIL-PRF-8625	SAE AS478	ASME B1.1
ASTM B211/B211M	SAE AMS4121	SAE AS8879	SAE AS3071
ASTM E1417/E1417M	SAE AMS4124		

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 11
DLA - CC

Preparing activity:
Navy - AS

(Project 4730-2024-001)

Review activities:
Army - AT
Navy - CG

Industry Association:
SAE E-25

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can changes, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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